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ORIGINAL PAPER

Applying and benchmarking a stochastic programming-based bidding strategy for day-ahead hydropower scheduling

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Abstract

Aneo is one of the first Nordic power companies to apply stochastic programming for day-ahead bidding of hydropower. This paper describes our experiences in implementing, testing, and operating a stochastic programming-based bidding method aimed at setting up an automated process for day-ahead bidding. The implementation process has faced challenges such as generating price scenarios for the optimization model, post-processing optimization results to create feasible and understandable bids, and technically integrating these into operational systems. Additionally, comparing the bids from the new stochastic-based method to the existing operator-determined bids has been challenging, which is crucial for building trust in new procedures. Our solution is a rolling horizon comparison, benchmarking the performance of the bidding methods over consecutive two-week periods. Our benchmarking results show that the stochastic method can replicate the current operator-determined bidding strategy. However, additional work is needed before we can fully automate the stochastic bidding setup, particularly in addressing inflow uncertainty and managing special constraints on our watercourses.

Keywords Hydroelectric power · Bidding · Benchmarking · Stochastic optimization

1 Introduction

For power producers, the physical day-ahead spot market is the primary reference for their output. This paper details Aneo's efforts to improve the day-ahead power sales process by transitioning from a traditional operator-determined bidding strategy to a new stochastic programming-based method. Aneo is a hydropower producer with decades of experience in electricity market operations, currently operating 2.6 TWh of hydropower annually in the Nordic market.

Extended author information available on the last page of the article

There is a large body of academic literature on the bidding problem, which refers to the challenges faced by power producers in offering their output to short-term physical markets for electricity, and by large demand-side users in bidding their required consumption. For reviews on the bidding problem, see for instance Steeger et al. (2014); Aasgård et al. (2019); Peng et al. (2023). Wozabal and Rameseder (2020) and Kim et al. (2022) report on related industrial applications for virtual power plants. We contribute to this literature by sharing our experiences with industrial implementation, benchmarking, and the use of a stochastic programming-based optimization model for hydropower bidding.

The existing method for hydropower bidding in Aneo relies on SHOP, which is an optimization model for short-term hydropower scheduling developed by SINTEF Energy Research (Skjelbred 2019). To create bids, SHOP is run as a deterministic model with forecasted prices and inflow as input, in order to create production schedules from which the bids are further constructed by the operator on duty. In addition, the operator often makes additional manual adjustments to either the model setup or the results, in order to mitigate known limitations in the current modelling framework. The adjustments are most often related to safety measures in the face of uncertain input parameters (price and inflow are only forecasted values when bidding is performed the day ahead of operations) or special considerations for the watercourses that are not properly modelled in the linear programming framework in SHOP or in our implementation of SHOP for our watercourse systems. In this work, we will refer to the current practice of bid generation as operator-determined bidding due to the fact that the skills and experience of the operators are such an integral part of the process.

The operator-determined method of creating bids is labor-intensive and requires mature experience and therefore in-depth training. In addition, the move towards higher time resolution in the power markets will increase the frequency of decisions to be made each day. This, along with the increasing variations in electricity prices the last couple of years, has revealed the need to move on to a more automated and model-based method for creating bids.

Fortunately, stochastic modelling of prices and inflow has been included in SHOP (Belsnes et al. 2016; Aasgård et al. 2018). With stochastic prices as input, SHOP is able to generate bid curves as a direct result of the optimization. The basis for this development include Fleten and Pettersen (2005), who introduced the modeling of piecewise linear bid curves into hourly electricity markets, and Fleten and Kristoffersen (2007), who put this into the context of (hydro) power production with uncertainty in prices and inflow. The current implementation of bidding in SHOP is based on the adaptation described in Aasgård et al. (2018) where scenario prices instead of exogenous price points are used to determine the breakpoints of the bid curves, and the bid restrictions themselves are reformulated as cross-scenario inequality constraints as explained in Aasgård and Skjelbred (2020).

A relevant strand of literature concerns benchmarking of stochastic programming models used in short term power operations. In the context of wind power, Staid et al. (2017), Sari and Ryan (2018), Wozabal and Rameseder (2020) and Rachunok et al. (2020) benchmark energy scheduling problems.

Aneo is the first hydropower company in the Nordics to take the stochastic version of SHOP into use for bid generation. In this paper, we will refer to the bids generated from the stochastic version of SHOP as stochastic-based bids. However, we should note that we do not in fact use the bids generated from SHOP directly, but that we rather use the results from a stochastic run of SHOP to create production schedules from which we build our own bid curves. As will be explained in Sect. 2, we do this to ensure high-quality, feasible and explainable bids.

When moving to a stochastic-based bidding method, the process of generating price input in the form of scenario trees also becomes an integral part of the total process. In the operator-determined bids, the uncertainty of the price forecast is tackled by the experience of the operators. In a more automated process however, price input and the optimization model are more interlinked. As will be further detailed in Sect. 3, we have therefore experimented with different ways of generating input to the stochastic model, in order to see how important this is as a driver of the performance of the stochastic bid model. Note that even though both uncertainty of price and inflow can be incorporated into SHOP, we only consider price uncertainty in this work, since this is a prerequisite of the generation of bids in the current implementation in SHOP (Aasgård et al. 2018). Uncertainty in inflow may be added to our modelling setup later, but for this first test we wanted to limit the complexity of the modelling both for explainability of the results and for keeping calculation times within the time limits set by current operational processes.

In total, this paper details our experiences with benchmarking and operational use of stochastic programming for short-term hydropower bidding. In addition, we show results for one of our hydropower portfolios in a case study, as well as discussing some qualitative experiences we had with real-time testing of the new method.

2 Adaptations to the basic bidding method implemented in SHOP

The primary constraint implemented in SHOP to generate bid curves is that production volumes must be non-decreasing as prices increase. This is implemented in SHOP as cross-scenario inequality constraints, as detailed in Aasgård and Skjelbred (2020), ensuring that for each hour, the sum of production from the hydropower system is equal to or higher if the price in the current scenario is higher than in other scenarios. Enforcing this constraint allows for sorting hourly production volumes by price across scenarios to generate a non-decreasing bid curve. This produces a bid curve with columns corresponding to all potential prices for the bidding day, and rows for each hour. Production levels in the bid matrix are completed by interpolation between the optimized non-decreasing production volumes. Depending on the number of price scenarios, the bid matrix generated from SHOP may have more columns than allowed by the market operator. The default bidding method in SHOP therefore includes a method for reducing the bid matrix to the size allowed by the market operator (previously 64 but now 200 columns). In SHOP, the reduction method is based on a greedy heuristic that removes the least significant price columns after the bid matrix is optimized (Aasgård et al. 2018).

At Aneo, we have adopted the same principles as implemented in SHOP, but we compile and reduce our own bid matrix based on the optimization results. Specifically, we run SHOP with price scenarios and constraints enforcing non-decreasing production. Instead of using the output bid matrix bids directly, we have implemented our own method to generate bid curves from the optimized SHOP production results. This involves adjusting the bids for infeasible production levels in both single plants and cascaded systems, must-run production (due to minimum flow or other constraints), considerations for negative prices and addressing various watercourse-specific constraints. By using our own bid algorithm, we can better check and validate the optimization model results, ensure the feasibility of results for the physical production system and better track information potentially lost in the reduction algorithm.

We have also defined our own reduction algorithm to replace the internal method for this in SHOP. Our reduction algorithm also uses a greedy heuristic, but we have added extra logic to avoid reducing columns that are critical for the feasibility of the bids. “Critical" refers to columns of the bid matrix that have a larger influence of the shape of the bid curve. Examples of such columns are those reflecting the minimum production level, best-point operation, or shifting from one generating unit to two units. The criticality of columns in the bid matrix is measured using geometrical shapes similarly to (Aasgård et al. 2018), but with added weight/cost for some columns due to the aforementioned factors.

3 Price input and scenario generation

We have tested two different sets of price scenarios as input to the stochastic model. One set consists of ensemble forecasts from an external provider, while the other set is generated using an in-house-developed method. As part of the benchmarking setup, which will be explained in the next section, we compare these two sets of price inputs by creating stochastic bids with each. These stochastic methods are then compared to the deterministic price input used in the operator-based bids. While detailed descriptions of how to generate price scenarios are beyond the scope of this paper, we provide a high-level explanation of these two methods here.

The first set of price scenarios is obtained from an external third-party provider and is based on ensemble weather forecasts processed through an optimization/simulation model that considers the total power system. The external provider updates the price scenario output daily. These price scenarios largely maintain the profile one expects from the spot price. Power market prices usually have a daily profile with a peak in the morning and afternoon hours. Our hypothesis is that maintaining this profile may be advantageous for the resulting bids and production schedules, specifically related to start/stop of units as well as ramping restrictions, and pressure head optimization.

The second set of price scenarios is generated in-house using a deterministic price forecast with added errors drawn from a distribution of historically observed errors. The deterministic forecast used as the starting point for the in-house method is the same one that operators use as the basis for their decisions.

The error distribution is based on the historical differences between the main deterministic forecast and the realized market prices. The in-house-generated scenarios often do not retain the expected profile of the main deterministic price forecast and lack the underlying weather information present in the ensemble forecasts. However, the in-house scenarios often span a larger range of prices, which is important to cover the full operating range of the power plants in the bid curves. Examples of generated scenarios from each of these methods for a specific day are shown in Fig. 1.

When we test the different bid methods within the rolling horizon framework (explained in the next section), we assess the combined performance of the optimization model (the bid methodology) and the price scenario input, since these are very closely linked in a stochastic model. By including two alternatives for price input, we aim to identify where the greatest potential for further improvement lies—in the optimization model itself or in the quality of the input.

To further compare the alternative sets of scenario inputs, we calculate metrics on the scenario sets, specifically the energy score as defined by Gneiting and Raftery (2007) and the integrated distance as defined by Staid et al. (2017). Both metrics measure the distance between observed values and scenario values.

The energy score (see applications in, e.g. Gneiting et al. (2008) and Staid et al. (2017)) measures both the accuracy and spread of scenario sets. Mathematically, it can be expressed as follows:

$$\text{Energy Score} = \frac{1}{N}\sum_{i=1}^{N} P_i \|S_i - O\| - \frac{1}{2N^2}\sum_{i=1}^{N}\sum_{j=1}^{N} P_i P_j \|S_i - S_j\| \tag{1}$$

where N is the number of scenarios, S_i and S_j are the scenarios i and j of the scenario set, P_i and P_j are the probabilities of the i-th and j-th scenarios, O are the observed values, and $\|\cdot\|$ denotes the Euclidean norm. The energy score sums the weighted differences between observed and scenario values, then subtracts the weighted differences between all pairs of scenarios. The first term evaluates accuracy, measuring how well scenarios match observed data, while the second term evaluates spread, penalizing overly diverse scenarios. A lower energy score indicates better forecast performance. When applied to deterministic forecasts, the metric reduces to the accuracy component alone.

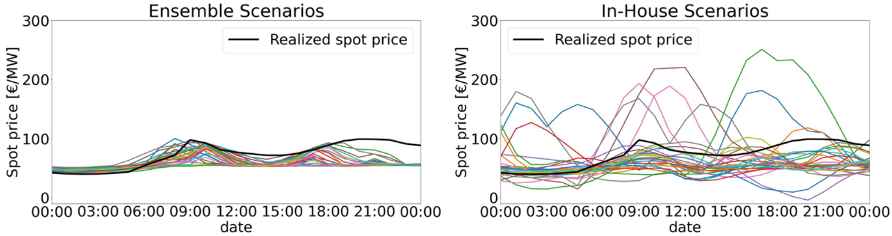

Fig. 1 Example of generated scenarios for the bidding day, using ensembles (left) and the in-house method (right). The realized market price for this day is shown as a thick black line

Integrated distance (Staid et al. 2017) is calculated as the expected value of the absolute differences between the observed values and the scenario values, each weighted by the probability of the corresponding scenario. Mathematically, it can be expressed as follows:

$$\text{Integrated Distance} = \sum_{i=1}^{N} P_i \sum_{j=1}^{M} \left| s_{i,j} - o_j \right| \tag{2}$$

where N is the number of scenarios, M is the number of observed values, o_j is the observed value at the j-th time step, $s_{i,j}$ is the value of the i-th scenario at the j-th time step or instance, and P_i is the probability of the i-th scenario. While the metrics for energy score and integrated distance allow us to compare some important properties of the input to the optimization model, we believe that the combined performance of the optimization model and the input, tested in the benchmarking framework, is the most crucial for establishing trust in the new bidding method.

Finally, the number of scenarios used to represent the sample space for possible future prices significantly impacts the calculation time of the stochastic models. Ideally, we would like to use a large number of scenarios, but this is not possible in terms of calculation time within the current time limits set by the market operator and our internal processes. In our current setup, 25 scenarios are used for both the sets of stochastic input.

Concerning the scenario tree structure, we use a fan tree that branches before the bidding day. According to the recommendations in Belsnes et al. (2016), we add both maximum and minimum scenarios to cover the full operating range of the production system in the bids. We generate these extreme scenarios by multiplying the main deterministic forecast with given weights and assign them a very low probability to avoid distorting the original price expectation.

4 Benchmarking methodology

To evaluate the performance of the stochastic-based bids, we wanted to compare it to our current process based on operator-determined bids. Implementing this setup presents several challenges. First, we do not compare the bids or bid matrices themselves directly. Rather, we aim to measure the consequences of the bids, i.e., the resulting production schedules after market clearing. Since we cannot send multiple alternative bids to the market, we constructed our own method to test the bids each day. We do this by creating a counterfactual experiment where we generate parallel bids each day using both the operator-determined bids (which are sent to the real market) and the stochastic-based bids (which are not sent to the real market but are evaluated based on the realized market prices). This comparison is fair because we act as a price-taker in the markets, i.e., we act as if our bid strategy does not affect the resulting market price.

For a detailed explanation of our experiment, please see Fig. 2. The top part of the figure shows the current operational process using operator-determined bids, while the bottom part illustrates the counterfactual setup. The differences

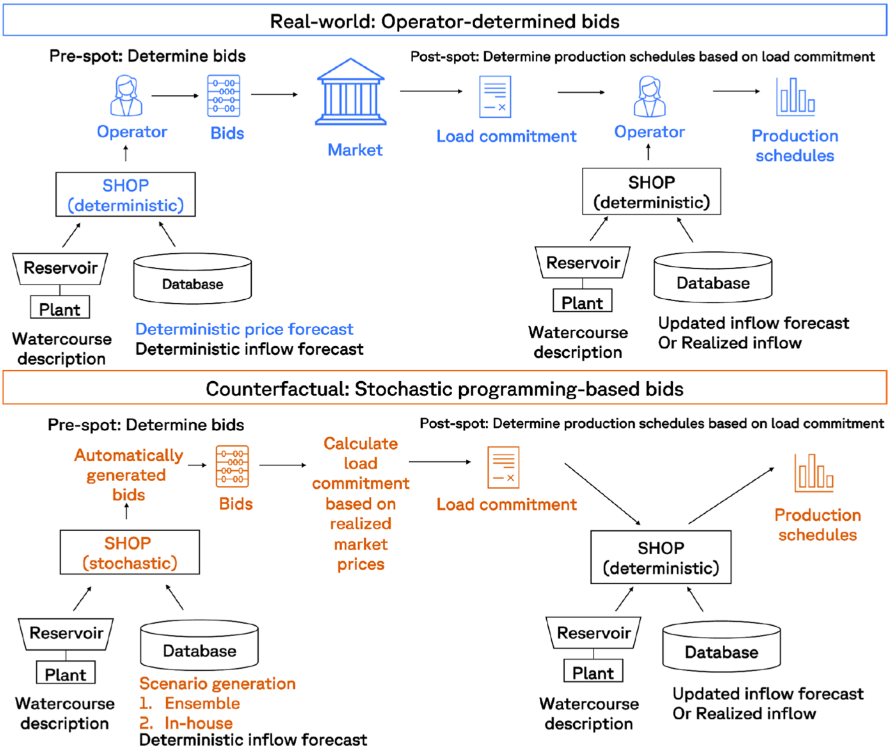

Fig. 2 Illustration of the process for the real-world setting and the counterfactual experiment for the stochastic models

between the two methods are indicated by the colors, blue and orange. Beginning at the top of the figure, the daily process using operator-determined bids is as follows. Before the day-ahead spot auction, the operator uses SHOP in deterministic mode to create bids based on the watercourse descriptions and information from our database. The database contains the deterministic price and inflow forecasts for the current optimization period, as well as any time-dependent constraints for the optimization model. The bids are sent to the market, and after the auction ends, we receive a load commitment for the next day, as well as realized market prices. The load commitment is the total sum of production for our entire portfolio of hydropower plants and other production assets, and it must be allocated to individual production units. The operator then uses SHOP again to generate optimal production schedules for each power plant that adhere to the market commitments. In this post-spot step, we use updated inflow forecasts from the database and realized inflow values since these are now available for the time that has passed since the bids were submitted.

Moving to the lower part of the figure, we observe how the parallel counterfactual experiment is set up for the stochastic method. In this method, there is no operator in either the pre-spot or post-spot phase. Instead, SHOP generates the basis for determining the bids in stochastic mode using the same watercourse description and time-dependent constraints from the database. Price input in the form of scenario trees is obtained either from ensemble forecasts or our in-house developed price methodology. The counterfactual experiment also uses the same deterministic inflow forecasts as the operational setting. The stochastic-based bids are not sent to the real market, but we use the realized market prices to interpolate the bid curves and obtain our counterfactual load commitments. SHOP is then re-run to determine the production schedules, this time without any operator-interaction for the post-spot run.

This experimental setup, particularly using the same watercourse models, constraints and inflow forecast for all our runs, ensures that we are able to evaluate the performance of the stochastic bids thoroughly in real cases and in an operationally realistic setting. The setup for our counterfactual experiment also lays the groundwork for building a fully automated framework for day-ahead bidding of hydropower.

To evaluate the long-term effects of using automated, stochastic-based bids compared to the existing operator-determined bids, we repeat the counterfactual experiment described above for each day of our test period in a rolling manner, as illustrated in Fig. 3. Specifically, each day begins with the schedule resulting from the previous day's bid. We calculate counterfactual starting reservoir levels by considering the discharge from the counterfactual production schedule and updating them based on inflow forecasts or actual inflow, depending on the information available. We then calculate new bids for the upcoming bidding day, which, after market clearing, results in a schedule applied at the start of the following day. This setup is the same for both the user-operated bids and the stochastic cases.

However, there are some limitations, particularly related to the reservoir management strategy for periods longer than the usual two-week time horizon

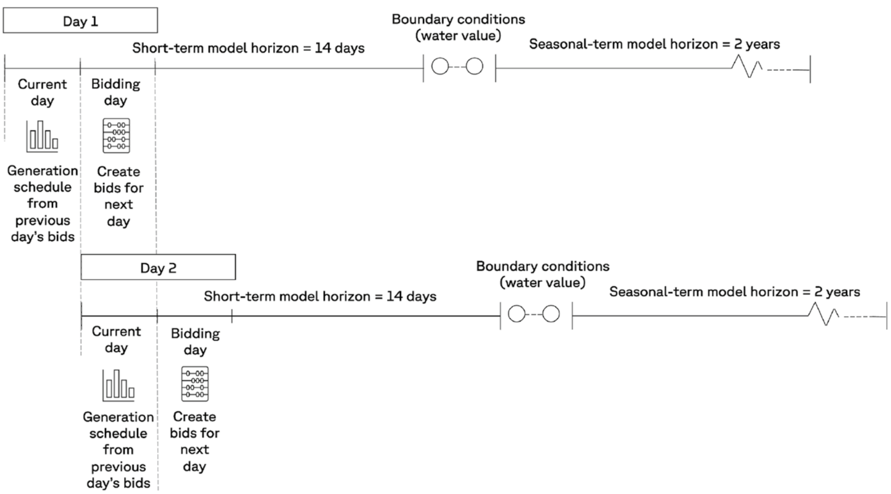

Fig. 3 Illustration of our rolling benchmark setup that repeats both the pre-spot and post-spot process of bidding and scheduling each day

covered by SHOP. For all our runs with SHOP (both the deterministic and stochastic version), we require boundary conditions to ensure a reasonable reservoir management strategy over time. This concept is further illustrated in Fig. 3 which indicates the model horizons. In our operational setting, boundary conditions are established by coupling with a seasonal model that uses an SDDP-based method to calculate the reservoir strategy for the next two years. We use SINTEF's ProdRisk model for this (Gjelsvik et al. 2010) which provides a cut-based description of the value of water at the end of the short-term horizon. Details of the seasonal scheduling method are beyond the scope of this paper; however, in essence we are able to connect our market and scheduling decisions in SHOP to an optimized longer-term reservoir management strategy.

However, due to resource limitations, we cannot set up a full parallel run of the seasonal model for the counterfactual experiment. Instead, we use the same operationally determined end-of-horizon water value descriptions for both the stochastic and operator-determined cases. Although the end-description from ProdRisk is quite flexible, it might not provide the right incentives to the short-term model in our counterfactuals if the decisions taken within the short-term horizon significantly diverge from the operator-determined strategy. If we let the rolling benchmark framework run for extended periods, the greater is the chance that we end up in inconsistent situations for the counterfactual stochastic cases. To mitigate this, we run our experiment for two weeks at a time, resetting the initial conditions (i.e., the start reservoir levels) for the stochastic cases at the end of each two-week period. This ensures more accurate end-conditions but also means we are comparing consecutive two-week periods rather than individual days over longer durations. We believe that this is a reasonable trade-off between short- and long-term effects of the bidding strategies.

To actually compare our results, our test framework logs the outcomes of applying the production schedules from the after-spot runs to the realized prices and inflows of that day. This way, we measure the consequences of the bids, not the bids themselves. What we measure is the daily revenues from the market (i.e., market price times the produced volume), the start-up costs, and the change in the value of water left in the reservoirs after the bidding day. The latter indicates the cost (in terms of water usage) of producing the load on any given day. In total, our daily objective function (that we term "daily grand total") consists of the value of the revenues minus the start-up costs and the change in reservoir value as shown in Eq. 3). The daily grand total and each of its constituents are calculated daily, and we also aggregate the numbers per two-week period.

$$\text{Daily grand total} = \text{revenues} - \text{costs} - \text{change in reservoir value} \tag{3}$$

We need to explain in more detail how we calculate the change in the value of water left in storage after the bidding day. The value of water (or water value) is an important concept in hydropower scheduling because it represents the marginal cost of production. The water value can be seen as the alternative or opportunity cost of using one unit of water for production now, which could have been saved for later when prices might be higher. Referring again to Fig. 3, water values are used for coupling between the short- and seasonal-term planning.

However, in our rolling framework, which mainly considers the short-term horizon, we focus on measuring the value of water used for production *on the bidding day* rather than the entire scheduling horizon. Therefore, we use the internal water value in SHOP as the measure of water value for production. SHOP has functionality to output the internal (i.e., within the short-term horizon of two weeks) value of water as time series from the optimization model. These internal water values are in fact the shadow prices of the reservoir balance constraints for each reservoir modelled in SHOP, indicating the worth of one additional unit of water for each time step. More details can be found in Skjelbred (2019). In our setup, we calculate the value of water used for production on the bidding day as follows:

$$\text{Change in reservoir value} = s_{\text{end}}\lambda_{\text{end}} - s_{\text{start}}\lambda_{\text{start}} \tag{4}$$

that is, the storage at the end of the bidding day (s_{end}) times the shadow price (λ_{end}), minus the storage at the start of the bidding day (s_{start}) times the shadow price (λ_{start}). This gives us a change in the value of water in storage occurring from the decisions on production made during the bidding day only.

5 Case study and results

5.1 Aggregated results and grand total

We run the rolling horizon benchmark for one of our watercourse portfolios located in the NO3 price area in Norway. Figure 4 illustrates the watercourses and their key characteristics. The portfolio consists of both simple one-reservoir,

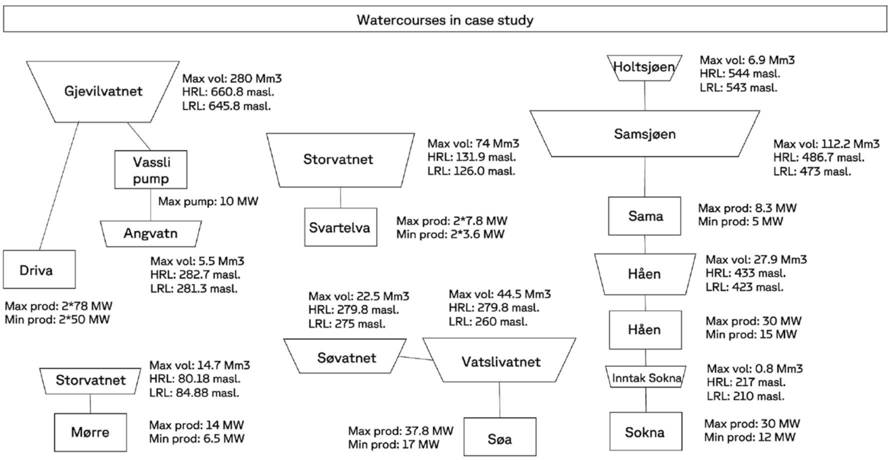

Fig. 4 Watercourses tested in the rolling horizon benchmark

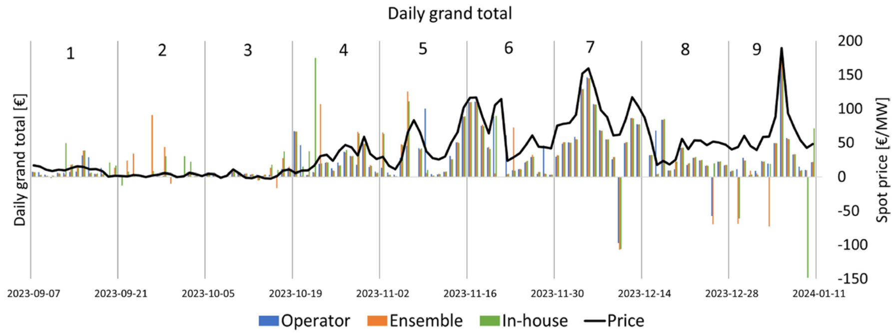

Fig. 5 Daily grand total for all days tested in the rolling horizon benchmark, including the daily average realized price. The x-axis labels indicate the start dates for each two-week period

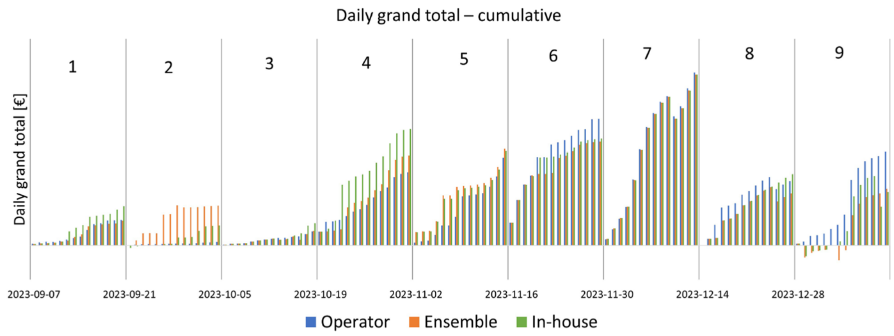

Fig. 6 Daily grand total for all days tested in the rolling horizon benchmark, cumulative by period

one-plant systems as well as more complex cascades. In our operational scheduling, we optimize these watercourses together as one portfolio.

In the rolling horizon benchmark, we run three different cases: operator-determined bids that correspond to our actual market bidding, and then the stochastic-based bid method with two sets of price input: ensemble forecasts from an external provider and our in-house developed method. Our tests run for approximately four months, from September 7, 2023, to January 10, 2024. As explained above, we compare sets of consecutive two-week periods due to the end-value setting for the counterfactual.

Figure 5 shows the daily grand total calculated for each day of our test period, along with the daily average realized price. Figure 6 shows the daily totals accumulated over each two-week period. In both plots, we have removed some days (specifically November 7, 2023, November 21, 2023, and December 14, 2023) because the results showed very high spikes that distorted the representation of the other periods. This will be further explained in Sect. 5.5.

Table 1 shows the relative difference between the sum of the daily totals for each period with respect to the absolute value of the operator result:

Table 1 Daily grand total: presented as the relative change with respect to the operator result, aggregated per two-week period. In the final row, Equation (5) is applied to the sum of the nine periods

Period	Operator	Ensemble	In-house
1	0	− 0.04	0.53
2	0	10.43	4.72
3	0	0.03	0.58
4	0	0.23	0.60
5	0	0.54	1.06
6	0	− 0.02	− 0.01
7	0	− 0.01	− 0.01
8	0	1.02	0.00
9	0	− 0.39	− 0.34
Whole test period	0	0.53	0.27

$$\frac{S-D}{|D|} \tag{5}$$

where S is the result of the stochastic-based model with each price input, and D is the operator-determined result. Note that the daily grand totals may be both positive and negative, due to the change in the value of water in storage. Consequently, the results in the table may exceed 1 in some periods. Higher positive values indicate better results.

From these overall results, it appears that the stochastic-based models perform best. They do well in the first five periods, while the operator-determined bids perform better in the latter. The total sum of the daily grand totals is reported on the last row of Table 1. It shows that the stochastic model with ensemble prices achieves the highest result, being 53% better than the operator-determined, followed by the model with in-house prices, which is 27% better than the operator-determined model. However, it is necessary to dig deeper by analyzing the individual components that make up the daily grand total, i.e. the daily revenues, the start-up cost and the change in reservoir value.

5.2 Revenues

Figure 7 shows the revenues obtained for each bidding day of our test period, as well as the realized (daily averaged) market prices in the same period. Table 2 shows the sum of revenues in each period for each method in relative difference with respect to the absolute value of the operator-determined result. Revenues will always be positive in our setup, unless we have forced production at negative prices due to, for instance, minimum flow constraints. There are some hours with negative prices during our test period, but these instances are so rare that they do not affect the aggregated daily results.

Figure 7 shows that the revenues are higher in the periods where the prices are higher, i.e. for periods 4–9. This is intuitive because, even if we produce the same volume in all periods, higher prices would result in higher revenues. However, the

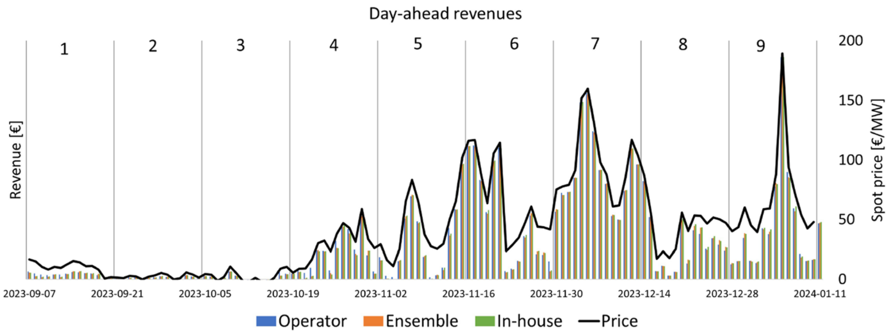

Fig. 7 Day-ahead revenues obtained for all the days tested

Table 2 Revenues: presented as the relative change with respect to the operator result, aggregated per two-week period

Period	Operator	Ensemble	In-house
1	0	− 0.12	− 0.08
2	0	− 0.08	− 0.04
3	0	− 0.12	− 0.11
4	0	− 0.10	− 0.09
5	0	− 0.03	− 0.02
6	0	− 0.02	0.00
7	0	0.00	0.00
8	0	0.04	0.06
9	0	0.00	0.01
Whole test period	0	− 0.01	0.00

differences are larger than just this price effect; this is because all our three model runs are able to schedule the production reasonably well in response to the price: higher volumes are offered to the market when prices are high compared to lower-priced periods. We also observe the same effect in the results for reservoir value, which will be discussed in Sect. 5.4. The values in Table 2 show that the stochastic models have slightly less revenue the first 5–6 periods and slightly more the latter periods, although the differences are small.

5.3 Start-up costs

We cannot assess the performance of the different models based on revenues alone. We also need to consider the cost aspect. In hydropower, the most important production costs are start-up costs and the cost of the water used for production. Start-up costs are related to water loss and equipment wear from starting and stopping the generating units. Results for start-up costs are shown in Fig. 8 and accumulated per period in Table 3 as a relative difference with respect to the absolute value of the operator's start-up costs. We report start-up costs as positive

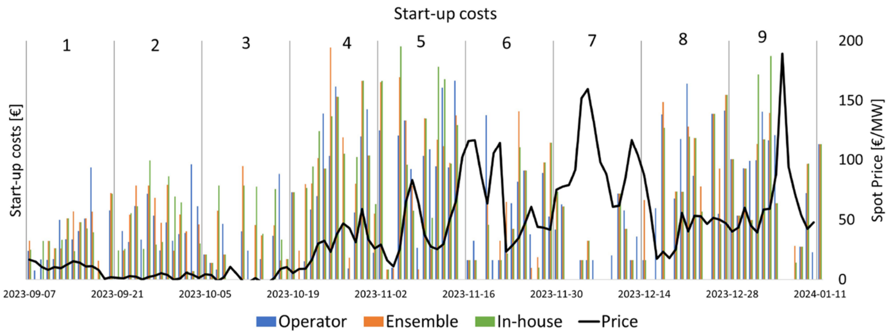

Fig. 8 Start-up costs for all the days tested

Table 3 Start-up costs: presented as relative change with respect to the operator result,aggregated per two-week period

Period	Operator	Ensemble	In-house
1	0	0.06	− 0.02
2	0	− 0.04	− 0.02
3	0	0.25	0.54
4	0	0.32	0.27
5	0	− 0.03	0.04
6	0	0.04	− 0.14
7	0	− 0.12	− 0.12
8	0	0.03	− 0.11
9	0	− 0.07	0.01
Whole test period	0	0.05	0.03

numbers here; however, note that, as they are cost elements, they are subtracted from the revenues in the calculations for the grand total. Figure 8 illustrates that there is a tendency for start-up costs to be lower in the periods with high prices. This is logical because most generating units would already be turned on in these periods, reducing the need for frequent starts and stops.

Table 3 shows that the stochastic runs have slightly higher overall start-up costs, with a significant increase in periods 3 and 4. Operators tend to select bids that result in “blocks” of production at the same level for consecutive hours, such as production during the day but not at night. This works well as a rule of thumb because prices are usually higher during the day. The daily aggregated prices shown in the plots do not fully reflect the hourly price variation within each day, but it appears that the stochastic bid methods are more willing to adjust production according to these variations, leading to increased start-up costs. In periods 3 and 4, where the start-up costs are higher, the stochastic runs achieve higher grand totals that the operator-determined bids, indicating that more flexible production patterns may have value. However, start-up costs are in general minor

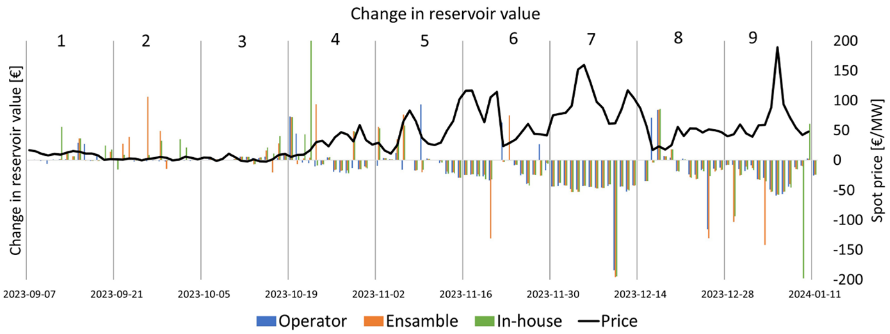

Fig. 9 Change in the value of water in storage during the bidding day for all days tested

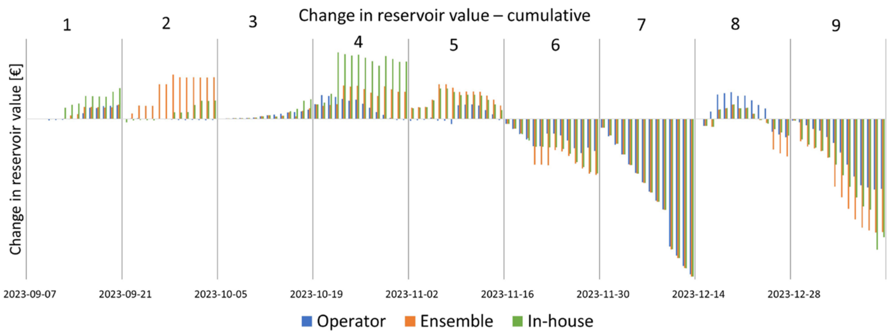

Fig. 10 Change in the value of water in storage during the bidding day for all days tested, cumulative per period

compared to the revenues obtained, so it is more important to investigate the main cost element for hydropower, which is the cost of water used for production.

5.4 Change in reservoir value

Figure 9 shows the calculated change in the value of water in storage for the bidding day across all tested days, with accumulated results per period shown in Fig. 10. Table 4 presents the sum per period. Since we report this as the *change* in the value of water in storage between the end and start of the bidding day, the values can be both positive and negative. This is also reflected in the grand totals being both positive and negative. A negative change represents a cost, indicating *less* value in storage at the end of the day than at the start. A positive change indicates *more* value in storage at the end of the day than at the start.

In general, our results for the change in the value of water reflects the production pattern over the period. In the first half, the values are mostly positive, indicating that we are saving water by not producing as much. Long-term water values, which we use as boundary conditions after the short-term horizon, are also increasing in

Table 4 Change in reservoir value: presented as relative change with respect to the operator result, aggregated per two-week period

Period	Operator	Ensemble	In-house
1	0	0.06	1.26
2	0	42.22	19.15
3	0	0.14	1.09
4	0	28.20	58.23
5	0	0.51	1.00
6	0	− 0.02	− 0.01
7	0	− 0.01	− 0.01
8	0	0.99	0.00
9	0	− 0.62	− 0.56
Whole test period	0	0.47	0.23

this period as we move towards the winter season. In the second half of our test period, prices and production levels are higher, and the results for change in reservoir value are mostly decreasing. Essentially, we transition from a filling season to a drawdown season during the test period, which is reflected in our results.

From the plots of reservoir value and the results in Table 4, we observe significant differences in the value of the reservoir during periods 2 and 4. In period 2, the stochastic method with ensembles deviates from the other two strategies, while in period 4, it is the in-house method that deviates. The revenues and start-up costs for these periods are similar across all three methods, indicating that the production commitments are also similar. The significant difference in reservoir value, and consequently the difference in grand total (particularly high in period 2), is therefore related to the water values used as boundary conditions for short-term optimization.

As explained in Sect. 4, we use the water values obtained from the operational setting also for the counterfactual experiments. The water value description is therefore better adapted to the operator-determined method, meaning the conditions will best represent the value of water at reservoir levels equal to or close to those reached by this method. The stochastic methods might end up at reservoir levels farther from the operator-determined levels, making the water value description less suitable for these cases. This results in a reservoir value that is sensitive to changes in reservoir levels, typically near the lowest or highest regulated water levels (i.e., edge cases for the coupling between hydropower models of different horizons). Additionally, some of the reservoirs in our portfolio are so small that they can be emptied and refilled within the short-term horizon, sometimes within just a few days. For these reservoirs, the inflow pattern and prices within the short-term horizon are the main drivers for the scheduling strategy, rather than the long-term water values.

5.5 Reservoir strategies

To illustrate the discussion on the value of water in storage and boundary conditions, consider Fig. 11 which shows the reservoir strategy for a large reservoir, Gjevilvatnet. Here we see that the strategies from all the models follow each other

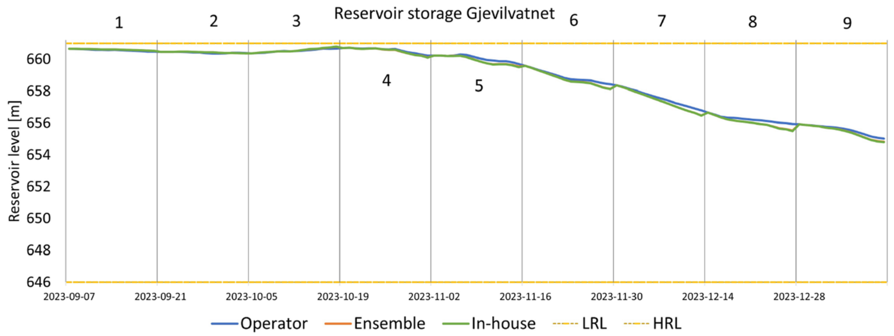

Fig. 11 Reservoir trajectories for Gjevilvatnet. The two stochastic cases overlap as the green line. Minimum and maximum allowed levels for the reservoir are indicated by the dashed orange lines

quite closely. For the last few periods, the stochastic models deviate slightly towards the end of each period, but the differences are not large enough to affect the water value description results. In Fig. 12 the situation is somewhat different. Storvatnet at Mørre is a smaller reservoir, and the stochastic models choose to manage the reservoir quite differently from the operators, which is allowed only within each two-week period. If this strategy deviates significantly from the operators' and the water value cut description is sensitive to changes in the reservoir level, we end up with a situation where the change in reservoir value is large even if the production schedules are quite similar, as explained above for period 2. In the plot for Storvatnet, we see that the reservoir strategies are quite different in periods 1, 2, 3, and 8. At the start of each new two-week period, the reservoirs are reset to the same level as obtained by the operators.

Another aspect shown in Fig. 12 is that the reservoir sometimes exceeds the highest regulated water level, most notably in period 3 for the operator-determined

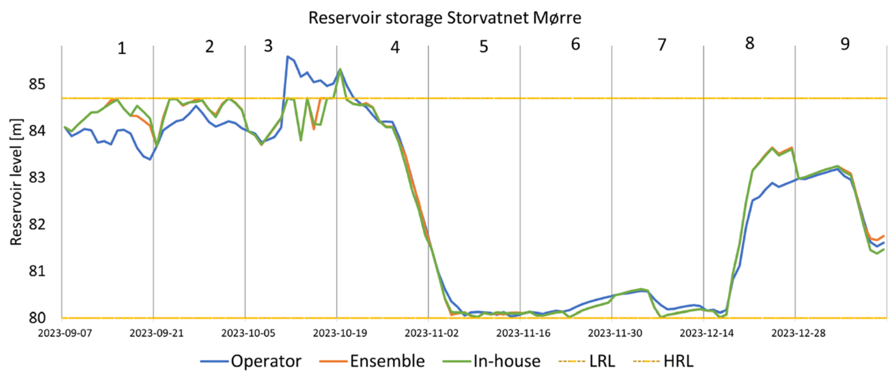

Fig. 12 Reservoir trajectories for Storvatnet Mørre. Here it is evident that the two stochastic methods choose a different reservoir management strategy within each two-week period but are adjusted back to the operator-determined/real-world case at the start of each new period

bids. This results in water spilling over the dam and into the river. It appears dramatic in the figure, but in reality, this occurs occasionally when inflows are large. This constitutes a loss of water that could have been saved and used for production if there had been more room in the reservoir, but it does not result in any broader losses or damages in the physical area where the reservoir is located. Both stochastic methods are able to save more of the water lost by the operator's strategy in this situation.

Another result not evident from the figure is that all three strategies actually go below the lowest regulated limit in periods 5 and 6, and the stochastic cases also do so in period 8. In this period, we had a strict self-imposed limit on the reservoir in order to change some measurement equipment at the site. All models recognized this constraint, and we began to draw down the reservoir at the end of period 4, keeping it at a very low level throughout periods 5 and 6. There were some moderate high price peaks in these periods, and the stochastic models chose to produce more than the operators at these peaks. This resulted in a few hours being below the lowest regulated limit. Referring back to how our rolling benchmarking framework is set up as shown in Fig. 3, it might be that the optimized bids and resulting schedules based on the inflow forecast would have been within the limits for the stochastic cases, whereas with the actual realized inflow, we violated the minimum constraint. These situations caused the large spikes in the reservoir value results mentioned in Sect. 5.1. I.e. when we break the reservoir limits as we do on November 7, 2023 (Period 5), November 21, 2023 (Period 6) and December 14, 2023 (Period 8), the internal short-term water value that we use for evaluation of the change in reservoir value will be influenced by penalty slack variables in the short-term optimization model, and thus does not adequately represent the value of using water. If this situation occurred in a real-life setting, we would have traded the production volumes in the intraday market to keep the reservoir within its limits. However, this is not accounted for in the rolling benchmark framework, resulting in outlier results for these days. The spikes are not plotted in the graphs but are included in the overall results in Table 4.

The operators are able to consider inflow uncertainty when they submit the bids, whereas our stochastic methods do not yet incorporate inflow uncertainty. For smaller reservoirs and situations where we impose strict restrictions, inflow uncertainty might be more important than price uncertainty, and the strategies should ideally be robust for both.

In general, we see that even though the overall results of daily grand total favor the stochastic runs, this result is in fact mainly due to the change in reservoir value. The stochastic runs obtain (slightly) less revenues, and higher start-up costs. As the method for reservoir valuation might be better adapted for the operator-determined runs, more work is needed before we can have a high-quality automated stochastic bidding setup. Including inflow uncertainty may be part of a solution towards more robust bids. We also need to make sure watercourse-specific considerations and trade-offs are handled with care, which will require close collaboration with the operators. Additionally, better bids may be achieved by making use of extended SHOP functionality to improve the modelling of the physical characteristics and dynamics of the watercourses.

5.6 Comparison between price inputs

As a final result, we also want to comment on the differences between the two stochastic runs, i.e., how the method used for generating scenarios influences the performance of the model. The runs with ensemble price scenarios showed a better performance in total, achieving an aggregated daily grand total of 0.53, compared to 0.27 for the in-house scenarios. This supports our initial hypothesis that the ensemble scenarios would give the best performance because they better represent the price profile. However, the results vary a lot from period to period.

To further investigate these results, we have calculated some statistics and evaluation metrics to compare the stochastic forecasts, shown in Table 5. The mean displayed in the first column of the table is the mean value of the weighted scenario mean over all days in our test period, and is compared to the mean of the deterministic forecast and the mean of the realized price. In the second column of the table, the mean standard deviation (SD) over all test days is displayed. This is the mean of the SDs calculated for each hour in the model horizon for each day. Standard deviation is not calculated for the deterministic forecast and realized price, because we are not interested in the SD between different hours but rather the variance covered by the scenario inputs.

As we can see, the three forecasting methods all have a higher expected mean than the realized price. The in-house scenarios have the mean closest to the realized, while also having the largest spread in terms of SD. This is interesting because the ensemble forecast has the most different mean, and a lower SD, while still performing best overall in the benchmarking. This indicates that other properties of the scenarios may also be important for performance, for example the shape, spread and accuracy of the price input which can be assessed through the energy score and integrated distance as explained in Section 3.

The results for these metrics for our scenario inputs are shown in the two last columns in Table 5. If we first compare the energy scores between the deterministic and stochastic forecasts, the deterministic forecast has a higher value than the stochastic alternatives. We take this as an indication that the accuracy of the current deterministic forecast is not good enough to rely on deterministic modelling, and that moving to stochastic forecasting methods and stochastic programming would be beneficial. This is consistent with our initial aim of implementing a stochastic programming method for the bidding problem. If we further compare the energy

Table 5 Comparing the different price forecasts. The values shown are the mean values over all days in our test period (from 2023-09-07 to 2024-01-10). Mean(standard deviation)

	(Weighted) mean	(Weighted) standard deviation	Energy score	Integrated distance
Ensemble scenarios	45.63(33.86)	21.71(15.24)	280.3(192.0)	4978(2660)
In-house scenarios	43.58(35.18)	31.25(12.42)	296.8(155.9)	6803(1567)
Deterministic forecast	44.02(32.40)	–	368.7(225.8)	4319(2410)
Realized price	42.96(42.06)	–	–	–

scores of the two stochastic inputs, the ensemble forecasts have smaller energy score than the in-house method, which is consistent with our benchmarking results where the ensemble has the best performance. However, the difference in energy score for the two sets of scenario input might not be significant. Also, Staid et al. (2017) mentions that energy score might not be a good measure for comparing scenario input with (visually) different characteristics, as is the case for our experiment. According to Staid et al. (2017), the integrated distance is better at distinguishing visually different scenarios. In our cases, the integrated distance is similar for the ensemble and deterministic inputs, while the value for the in-house forecast is higher. We take this as an indication that the ensemble and deterministic forecasts have similar characteristics in terms of “errors" or distance between the forecasted and realized values, while the in-house method tends to have larger forecasting errors. Our intention was for the in-house forecasting method to have larger variance, so this is somewhat expected. Using the metrics calculated here in further work to improve our forecasting models and defining scenario input might be useful, as it can help in screening new methods for further tests. However, since there are some limitations to both of the metrics, the best way to evaluate the performance of alternative scenario inputs for stochastic programming is to simulate using them in practice which is what we have done in the benchmarking experiment. This is also the conclusion drawn in Rachunok et al. (2020).

In addition to the analysis given here on price scenarios, we also note that we think that including inflow uncertainty would be the most significant improvement to our modelling setup as this is something that the operators currently take into account, and it is the main driver of the strategy for some of the smaller reservoirs. We therefore see including inflow uncertainty as a more important step towards automatic bidding than further improving the price scenario input.

6 Experiences from stochastic bidding in practice

In addition to running the rolling benchmarking framework to assess the performance of the new stochastic programming-based method, we have also made the bids generated from this method available to the operators in real time. This means that for a period, operators were able to see both the bids they generated themselves and the bids from the stochastic model side-by-side. Note that this was only done for the in-house scenarios. Operators then had the opportunity to choose if they wanted to use their own bid, or the bid generated from the stochastic model. Note that this was done at a different period than the rolling benchmark (specifically from June 2022 to March 2023, i.e., prior to the automated benchmarking framework as detailed above).

During the real-time testing period, we observed that operators predominantly chose their own bids. There may be multiple reasons for this, with the primary one being their preference for using familiar methods. Some of our operators have decades of experience, and very much feel the responsibility of getting things right in the market interaction process. Making errors or sub-optimal decisions in the power market can lead to significant losses for Aneo and, in the worst case, result in penalties from the market operator or other authorities. This responsibility, along with intimate knowledge of our watercourses, their dynamics, and current or historical limitations in modeling, may make the operators reluctant to try new methods—and rightly so in many cases. The main feedback we got from the operators was that they wanted control over the bids and the possibility to change model output to get things right according to their gut-feeling. During the first real-time test period of the stochastic method, the process for modifying the resulting bid matrix was cumbersome, leading most operators to prefer their usual process where they felt more in control.

In the existing method, the operators build the bids ‘bottom-up’, meaning that they consider each plant in every watercourse, and then create a bid that largely has consistent flow/discharge patterns. This approach is especially important for cascades or when balancing multiple conflicting constraints. The stochastic method generated bids that were perceived as more ‘top-down’, meaning that the operators found it quite hard to identify the production and flow patterns for individual plants from the resulting (often very large) bid matrix. Ensuring the bid’s behavior for individual plants was a primary reason for implementing our own method of constructing the bid matrix from the stochastic SHOP run instead of using the bid matrix directly. We wanted to be able to explain to the operators how the bid was generated from the plant production schedules, and the only practical way to do this was to have our own implementation that we could adapt according to special considerations for our watercourses. The operators all gave us very important input on which adaptations they saw as most important to include.

We also tried several other ways to mitigate the operator’s concerns, for instance different visualizations of how the production schedules would look for various price scenarios, or overview of the price levels that triggered production for each plant. All this mounted to a lot of new information that the operators had to process to compare the bids. With tight time limits in the market, there was often just not enough time to go through both their own method (the operator determined bids) and the stochastic bids before the final bids had to be sent to the market. In total, the operators ended up sometimes choosing the stochastic bids for the simpler systems (one-reservoir, one-plant) but almost never for the more complex cascades.

Even if the first round of real-time tests were not as successful, the operators also see a need for a more automated method for bidding, especially as the time resolution in the Nordic market is changing from hours to 15 min in the next couple of years. They have in general been welcoming to the idea of using a new method, but again, the responsibility of being the one who actually pushes the button to send the bids to the market, has demanded such an internalization of the quality control process that their gut-feeling is that the stochastic models were not ready for full automation.

However, working further with the automated benchmarking setup, we have uncovered and corrected several bugs in the stochastic modelling framework, and we are preparing for a new round of real-time testing with the operators. This will include better functionality for the operators to change the stochastic bids, more interaction between the development team and the operators, a full update of our watercourse models in SHOP, and including inflow uncertainty as input to the stochastic optimization in addition to price.

7 Summary

The main result from the benchmarking framework is that the stochastic bids can largely replicate the operator-determined bids. However, feedback from the operators during the real-time testing indicate that more work is needed on the automatic setup. Implementing a fully automated setup involves incorporating both price and inflow uncertainty into the optimization model, as well as improving the underlying watercourse models, better understanding of the coupling to longer term models, and gaining insight into the trade-offs between various considerations. We conclude that it is difficult to build a reliable autonomous method using tools that require extensive hands-on adaptation to achieve good performance. In our implementation, the underlying watercourse models used in SHOP do not sufficiently capture all the physical characteristics crucial for making good scheduling decisions. SHOP has functionality to mitigate some of these shortcomings, and in hindsight, it would have been wiser to update the watercourse models rather than adding additional complexity through price uncertainty to improve the bidding process.

Starting this path of implementing new, more complex modelling approaches has also made us realize even more how important our operators and human-computer interaction are to bring out the best of human experience and smart algorithms. As we move further towards a more automatic framework for hydropower bidding, ensuring that these human experiences are not lost but rather incorporated into the new method will be crucial for the success of any new practice.

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Declarations

Conflict of interest Non-financial interests: Author S.-E. F. is co-Editor-in-Chief of the journal.

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